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Flavoromic Exploration of Regional Variations, Consumer Preferences, and the Role of Volatile and Nonvolatile Compounds in Strong-Aroma Baijiu

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ABSTRACT: Regional strong-aroma baijiu was analyzed via SPME-GC-MS, derivatization-GC-MS, and GC-FID to explore the relationships among flavor characteristics, trace compounds, and consumer preferences. A total of 84 volatile compounds and 60 nonvolatile compounds were identified. The results showed significant differences in volatile compounds among samples from different regions. Jianghuai baijiu showed a higher alcohol content, while Sichuan baijiu was rich in esters, contributing to its intense flavor. Nonvolatile organic acids played a dual role in modulating the aroma-taste balance. They negatively correlated with the alcoholic and cellar aromas but, in cooperation with volatile acids, enhanced the sour aroma and sour taste. Additionally, they interacted with specific esters to improve the sweetness and taste complexity. Sensory evaluations, including quantitative descriptive analysis (QDA), Check-All-That-Apply (CATA), and hedonic tests, demonstrated that 1-pentanol and ethyl pentanoate were positively related to consumers' liking, whereas ethyl octanoate, ethyl decanoate, and ethyl 3-phenylpropionate showed negative correlations.

KEYWORDS: *strong-aroma baijiu, trace compounds, flavor, consumer preference, sensory analysis*

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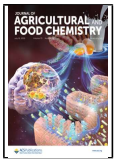
1. INTRODUCTION

Baijiu is a unique type of distilled spirit in China, with a long history and rich cultural connotations.[1] Its unique brewing process and flavor have made it one of the favorite alcoholic beverages among consumers, especially in China.[2] Sorghum, wheat, and other grains serve as raw materials for brewing baijiu through a traditional spontaneous fermentation. This process involves a series of steps: first preparing fermentation starters (qu), followed by grain cooking, controlled fermentation, distillation to extract the alcohol, careful aging, and final blending to achieve the desired flavor profile. At present, Chinese baijiu is primarily categorized into 4 main flavor types: strong aroma, sauce aroma, light aroma, and rice aroma. In addition to these basic aroma types, there are a variety of derived aroma types, which together constitute the diverse styles of Chinese baijiu, each with its own unique flavor and characteristics.[3,4] With its rich aroma, smooth taste, and remarkable cellar aroma, strong-aroma baijiu has become the most popular type of Chinese baijiu, accounting for about 70% of the total production of baijiu in China.[1]

The trace constituents in the strong-aroma baijiu determine its overall flavor profile, which is closely tied to factors such as geographical environment and brewing conditions.[5−9] Consequently, even among baijiu of the same strong aroma type, there are subtle variations in flavor characteristics from different regions.[10,11] Strong-aroma baijiu, based on geographical regions, can be divided into 3 categories: Sichuan category, Jianghuai category, and Northern category.[12] Currently, commercially available strong-aroma baijiu products are predominantly produced in the Sichuan and Jianghuai regions. This variation in flavor profiles stems from distinct brewing environments and production techniques. Sichuan-style baijiu exhibits intense cellar-derived aroma notes, whereas Jianghuai-style baijiu demonstrates more delicate and balanced aroma characteristics with refined mellowness.[12]

The Sichuan Basin's unique terroir (encompassing topography, subtropical climate, and purple soils) nurtures distinct baijiu flavor characteristics. Local glutinous red sorghum, thriving in mineral-rich terrain with over 90% amylopectin content, ensures both a high alcohol yield and full-bodied texture. This contrasts with Jianghuai-style baijiu, which utilizes amylose-rich sorghum from northern regions. Centuries-old cellars in Sichuan, with a wind-sheltered, humid climate, host exponentially expanding functional microbes. This environment shapes the intense, layered aroma of Sichuan baijiu. In the Jianghuai region, the high-temperature, humid climate also aids the brewing of microbes. But its flat terrain, dense water networks, and location in a climatic transition zone cause strong convective weather, reducing environmental microbe abundance; however, it helps fermentation to reach a new balance, creating the soft, elegant Jianghuai-style strong-aroma baijiu. Geographical and climatic differences thus lead to flavor variations between the 2 regions.

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According to previous reports, the number of trace compounds in strong-aroma baijiu has reached more than 2700, mainly including alcohols, aldehydes and acetals, esters, ketones, acids, aromatic compounds, nitrogen-containing compounds, sulfur-containing compounds, oxygen-containing heterocyclic compounds, terpenes, and phenols.[13] Currently, the focus of research on baijiu flavor compounds is mainly on volatile compounds as they are easier to detect and have a more apparent direct impact on the aroma profile of baijiu. However, the description of baijiu's flavor should not be restricted solely to olfactory characteristics but also to taste.[14] Recent studies have indicated that nonvolatile organic acids affected the volatile compounds in baijiu and may significantly influence the intensity of certain aromatic qualities.[15] Therefore, it is extremely necessary to conduct an in-depth exploration of the impact of nonvolatile compounds on the flavor characteristics of baijiu.

At present, sensory evaluation practices in the baijiu industry primarily rely on experts to maintain product quality and facilitate the development of new products.[16] However, consumer behavior drives the market valuation of baijiu, and its sensory characteristics are directly related to customer preferences. This dependency emphasizes the crucial importance of conducting a comprehensive investigation into the 3-way link among market trends, sensory qualities, and trace chemical components in baijiu production. Previous studies indicated that the trained panelists could distinguish the sensory characteristics of strong-aroma baijiu from different regions.[10] However, the flavor of strong-aroma baijiu has not been systematically studied, particularly from the perspective of consumer perception.

This study aims to achieve the following objectives: (1) systematically characterize both volatile and nonvolatile trace constituents in strong-aroma baijiu across different production regions; (2) identify nonvolatile compounds that are specific to certain regions, which help distinguish the geographical origins of strong-aroma baijiu; (3) explore regional differences using various sensory techniques, including quantitative descriptive analysis (QDA), Check-All-That-Apply (CATA), and hedonic tests; (4) demonstrate how specific nonvolatile acids directly influence consumer preference scores for flavor liking in strong-aroma baijiu. By synthesizing this information, the study seeks to investigate the chemical basis of sensory perceptions (aroma and taste) and differences in preferences.

2. MATERIALS AND METHODS

2.1. Baijiu Samples. Eight samples of strong-aroma baijiu were purchased from a local liquor store (Hubei, China). To minimize the analytical interference caused by alcohol, the samples were selected to have the same alcohol content level. The sample information was presented in Table 1. Prior to analysis, all samples were kept at 4 °C.

2.2. Analysis of High-Concentration Volatiles by GC-FID. In reference to previous studies with minor modification,[17] a 1 μL sample containing 1975.25 mg/L 4-octanol (internal standard) was injected into the GC-MS in splitless mode. The GC-FID analysis was performed using a 6890 gas chromatograph equipped with an HP-5MS column (30 m × 0.25 mm × 0.25 μm, Agilent Technologies, Santa Clara, CA). The oven temperature was initially set to 50 °C and maintained for 2 min. It was then increased to 80 °C at a rate of 1 °C/min and held for 5 min, followed by a ramp up to 250 °C at a rate of 20 °C/min, which was held for an additional 2 min. Nitrogen (N_2) with a purity of 99.999% was used as the carrier gas at a constant flow rate of 1.0 mL/min. Both the inlet and detector temperatures were set to 250 °C. Each sample was analyzed in triplicate. Compounds were quantified by using authentic standards; all standard curves showed good linearity, and the R^2 values were all above 0.99 (Table S1).

Table 1. Information on the Baijiu Samples

sample ID	group	ethanol content (v/v) (%)	origin	brewing method
JH1	JH	52	Suqian, Jiangsu, China	old five-pot order way
JH2	JH	52	Bozhou, Anhui, China	old five-pot order way
SC1	SC	52	Yibin, Sichuan, China	ferment in the different pit order way
SC2	SC	52	Yibin, Sichuan, China	ferment in the different pit order way
SC3	SC	52	Yibin, Sichuan, China	ferment in the different pit order way
SC4	SC	52	Yibin, Sichuan, China	ferment in the different pit order way
SC5	SC	52	Yibin, Sichuan, China	ferment in the different pit order way
SC6	SC	52	Yibin, Sichuan, China	ferment in the different pit order way

2.3. Analysis of Volatile Compounds by SPME-GC-MS. Volatile compounds of strong-aroma baijiu were analyzed by SPME-GC-MS, following the described method with slight modifications.[18] A 5 mL sample of diluted baijiu, with a final ethanol content of 10% (v/v), was placed in a 20 mL glass vial equipped with a silicon septum and saturated with 1.5 g of NaCl. A 10 μL portion of internal standard (4-octanol, 11,010 mg/L) was added. GC-MS analysis was performed using an 8890 gas chromatograph coupled with a 7000D mass spectrometer along with a PAL autosampler (Agilent Technologies, Santa Clara, CA) and an HP-5MS column (30 m × 0.25 mm × 0.25 μm, Agilent Technologies, Santa Clara, CA). Volatile compounds were extracted using a 2 cm, 50/30 μm DVB/CAR/PDMS fiber (Supelco Inc., Bellefonte, PA). The samples were equilibrated at 50 °C for 10 min and then extracted at the same temperature for 45 min while stirring at 250 rpm. The desorption time was 5 min.

The sample was injected using splitless mode. The heating procedure for the oven started at 50 °C for 2 min and then increased by 4 °C/min to reach 160 °C, where it was held for 5 min. The temperature was then raised at a rate of 4 °C/min to 230 °C and maintained for an additional 5 min. The transfer line temperature was set to 280 °C, and the ion source temperature was set to 250 °C. Helium was used as the carrier gas at a constant flow rate of 1 mL/min. Electron ionization (EI) mode was employed with an ionization energy of 70 eV. The identification of aroma compounds was performed in full scan mode, with a mass scan range set from 35 to 400 amu. The volatile compounds were identified by comparing their retention times and mass spectra to those of authentic standards or by searching the NIST 20 library.

Compounds were quantified by using authentic standards to create a standard curve; all standard curves showed good linearity, and the R^2 values were all above 0.99 (Table S1). A stock solution was prepared in a 52% ethanol/water (v/v) solution and then diluted to a range of concentrations to obtain a working standard solution. The compounds without authentic standards were semiquantified as relative to the concentration of the internal standard. Each sample was analyzed 3 times.

2.4. Odor Active Values (OAVs) of Volatile Compounds. The OAV of volatiles in baijiu was derived by comparing their concentrations to their thresholds in ethanol solution, and in general, compounds with an OAV ≥ 1 are considered significant contributors to the overall aroma profile.[19] The odor thresholds used in this study are derived from previous research.[13,20−23]

2.5. Qualification and Quantitation of Nonvolatile Compounds. The quantification of nonvolatiles was following the previous report with minor changes.[15] 10 mL of sample containing 1.05 mg/L salicin (internal standard) was concentrated to approximately 1 mL using a rotary evaporator and then transferred

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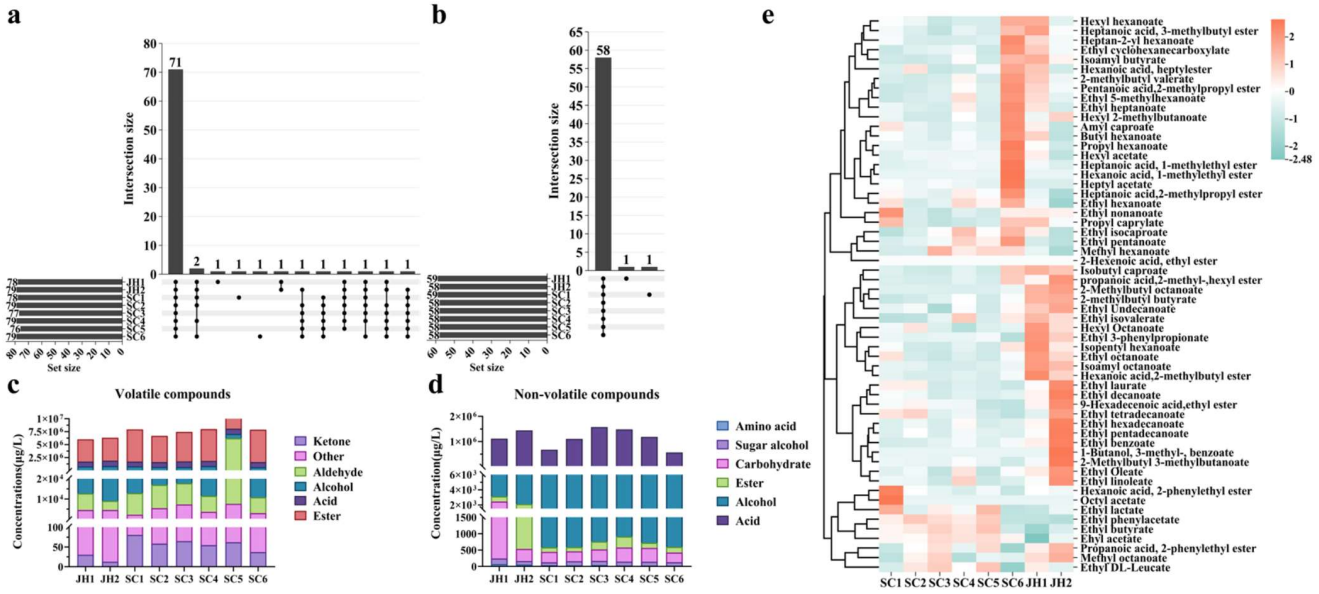


Figure 1. Comparison of the types and contents of trace components in different strong-aroma Baijiu. (a) Upset plot of the distribution of volatile compounds; (b) upset plot of the distribution of nonvolatile compounds; (c) concentration of the volatile compounds by category; (d) concentration of the nonvolatile compounds by category; (e) heatmap of esters in different baijiu samples.

to a 2 mL vial, which was subsequently dried with high-purity nitrogen gas. Lactic acid was measured separately because of its high concentration in baijiu. For lactic acid quantification, 100 μL of the baijiu sample was added to the 2 mL vial and completely evaporated.

For all dried samples, 100 μL of BSTFA (containing 10% TMCS) and 50 μL of pyridine were added to the prepared sample. The solution was then heated at 75 °C for 3 h, and 1 μL of the derivatized solution was injected into the GC-MS for analysis with a split ratio of 1:1. The oven temperature program was 65 °C for 2 min, increased by 6 °C/min to 280 °C for 5 min. Other instrument conditions were consistent with those described in 2.3.

All nonvolatile compounds were identified by calculating their retention indices (RI) and performing a preliminary comparison with spectral libraries from NIST 20, followed by comparison with authentic standards. The standard curves were established using authentic standards, and the concentrations of the target nonvolatile compounds were determined from these calibration curves. All standard curves showed good linearity, and the R^2 values were all above 0.99 (Table S2).

2.6. Quantitative Descriptive Analysis. For quantitative descriptive analysis (QDA), the sensory panel consisted of 10 panelists (5 males and 5 females) recruited from Huazhong Agricultural University. Panelists were selected based on their olfactory sensitivity and the ability to describe the odor and were trained before the formal descriptive analysis. The panelists were trained for basic aroma detection and scale use. During the training sessions (6 sessions, 1 h for each session), the assessors were presented with samples of commercial strong-aroma baijiu that represented a wide range of sensory characteristics. Descriptors were derived from a consensus vocabulary developed during panel training. Ultimately, 10 aroma descriptors (alcoholic, baked, cellar, fruity, floral, grain, sour, sweet, grass, and sauce aroma) and 9 taste descriptors (spicy, mellow, refreshing, lasting, sweet, astringent, soft, harmonious, and sour) were selected to characterize the flavor profile of strong-aroma baijiu. Panelists were trained with physical references (Table S3) to anchor descriptors to concrete sensory stimuli.

In the formal test session, the samples were evaluated using a five-point scale with 1 being "weak" and 5 being "strong." The samples were coded using 3-digit random numbers and presented in a random order. All sensory experiments were conducted in the sensory laboratory of Huazhong Agricultural University that was designed according to ISO 8589, under artificial daylight and temperature control (22 °C). Participants were told they were being recruited for a study of the baijiu flavor. The study was approved by the HZAU Institutional Review Board (IRB).

2.7. Consumer Sensory Evaluation. Forty-five panelists from Huazhong Agricultural University were recruited to participate in the sensory evaluation. Prior to recruiting participants, background checks were conducted to ensure that all consumers were over 18 years old with baijiu consumption experience and possessed normal sensory abilities. Each participant signed an informed consent form and provided personal information, including their age, gender, and habits related to baijiu consumption and shopping. Participants were told they were being recruited for a study on the baijiu flavor. The study was approved by the HZAU IRB, as the protocol was deemed to be free of potential harm. Each and all subjects read, agreed to, and signed a written consent form, which was also reviewed and approved by the HZAU IRB.

The test was divided into 2 parts: the first part was a hedonic test using an HGLM scale to assess aroma, taste, and overall liking (with the top end of the scale representing extremely like, the bottom end representing extremely dislike, and the midpoint representing neither like nor dislike); the second part was a Check-All-That-Apply (CATA) questionnaire, which covered aroma and taste descriptors. The CATA descriptors included in this study were initially selected based on published sensory lexicons of strong-aroma baijiu. The descriptors were then validated through informal tasting sessions using untrained consumers to make sure that they could be easily understood by untrained consumers. Eight samples from different origins were presented to consumer panelists in random order, with at least 3 min between samples, and consumers were asked to rinse their palate between samples. The samples were served in clear 10 mL baijiu tasting glasses labeled with a random 3-digit number that corresponded to their questionnaire.

2.8. Statistical Analysis. SPSS 27.0 (SPSS Inc., Chicago, IL) was utilized to conduct one-way analysis of variance (ANOVA) and Pearson correlation. Principal component analysis (PCA) and partial least-squares discriminant analyses (PLS-DA) were conducted using SIMCA 14.1 software. Graphs were created by using GraphPad Prism 9 and Origin 2017 software.

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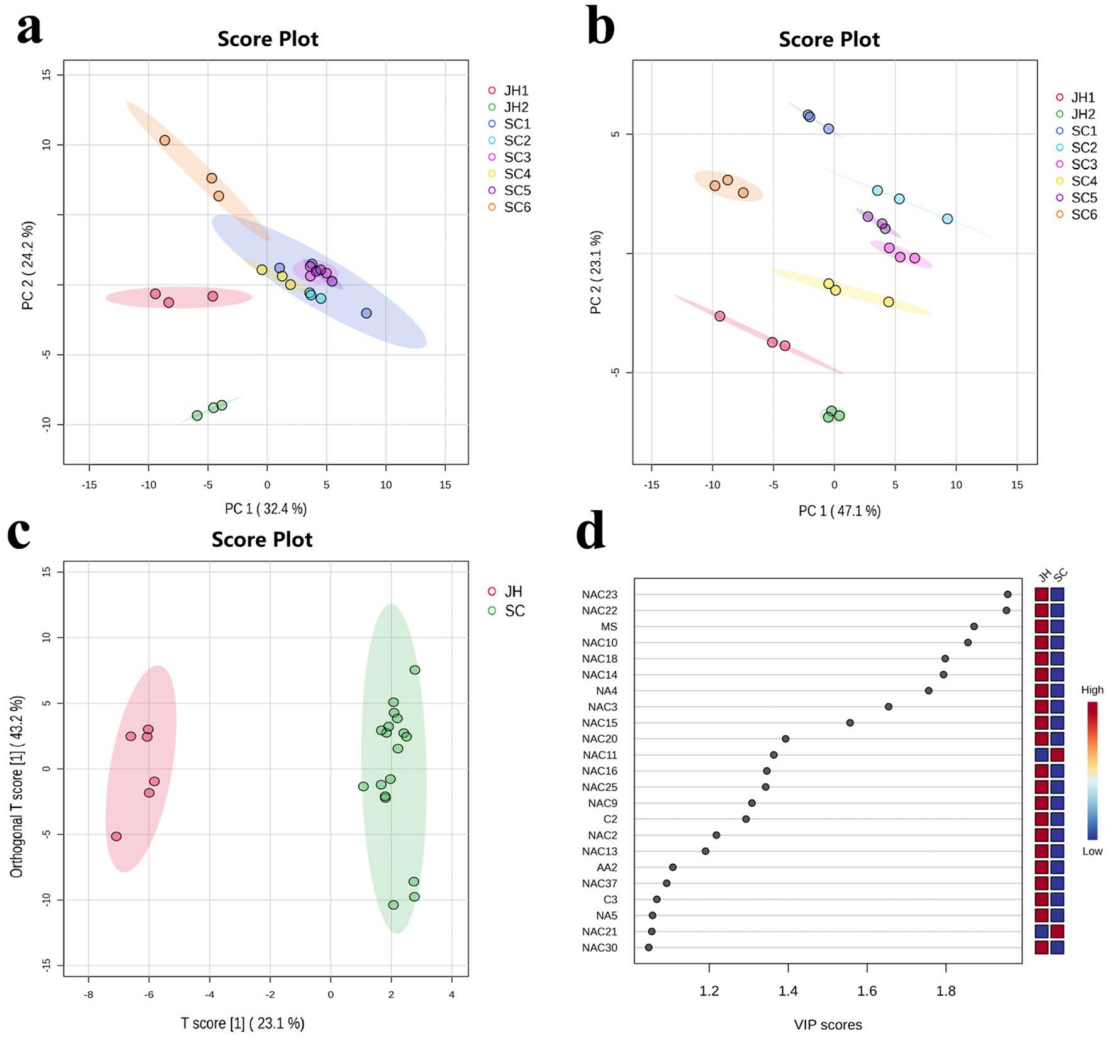


Figure 2. Multivariate statistical analysis of trace compounds of strong-aroma baijiu from different origins. (a) PCA of volatile compounds; (b) PCA of nonvolatile compounds; (c) OPLS-DA of "JH" and "SC" groups; (d) the bubble plot shows the compounds with VIP values >1 from OPLS-DA, while the heatmap shows the average concentration in the 2 groups. The compound ID could be found in Table 3.

3. RESULTS AND DISCUSSION

3.1. Volatile and Nonvolatile Compounds in Strong-Aroma Baijiu. In this study, a total of 84 volatile compounds were identified from strong-aroma baijiu, including 58 esters, 6 alcohols, 8 acids, 5 aldehydes, 2 ketones, and 5 other compounds (Table S1). As represented in Table S2, 60 nonvolatile compounds were confirmed, including 42 acids, 5 alcohols, 6 sugar alcohols, 5 sugars, 2 amino acid, and 1 ester. Overall, a total of 144 compounds were quantified. Esters are the predominant class of volatile compounds in baijiu, constituting a significant portion of the flavor profile. Meanwhile, organic acids constitute a significant component of the nonvolatile compounds in baijiu and exert a pivotal influence on the quality of baijiu. Specifically, isoamyl benzoate was detected only in sample JH2, octyl acetate and undecanedioic acid only in SC1, ethyl 2-hexanoate only in SC6, and alanine only in JH1. Meanwhile, decanoic acid was detected only in baijiu from Jianghuai regions, and heptyl acetate was only detected in baijiu from Sichuan regions. However, effective differentiation of strong-aroma baijiu cannot be reliably achieved through the presence or absence of a few individual compounds. The majority of these trace constituents were consistently detected across the same flavor type of baijiu, indicating a significant degree of compositional similarity among the trace constituents of strong-aroma baijiu (Figure 1a,b). This consistency makes it challenging to analyze the subtle differences in flavor within the same aroma type of baijiu on the basis solely of the types of trace constituents present.

3.2. Differences in the Quantity of Volatile Compounds among Various Strong-Aroma Baijiu. A total of 84 volatile compounds were quantified in strong-aroma baijiu (Table S4). As shown in Figure 1c, esters were found to be the most abundant compounds in strong-aroma baijiu, followed by short-chain organic acids and alcohols. Specifically, samples from the Jianghuai region have significantly higher alcohols than those from the Sichuan region, while samples from the Sichuan region have more esters. In the aroma composition of strong-aroma baijiu, esters are the most abundant components and play a crucial role in the aroma characteristics of the liquor body, imparting a rich fruity and ester aroma to the liquor. Therefore, heat maps and hierarchical clustering analysis (HCA) were used to analyze the ester contents of baijiu from different regions (Figure 1e). The results showed that

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Table 2. Aroma Compounds with OAVs > 1 in Different Strong-Aroma Baijiu

			OAV							
ID	compound	odor threshold (μg/L)	JH1	JH2	SC1	SC2	SC3	SC4	SC5	SC6
A1	1-propanol	54	2.23	2.68	3.70	3.18	2.81	3.93	3.58	3.59
A2	isobutanol	40	4.55	2.25	1.85	1.69	1.71	1.90	1.90	2.00
A3	1-butanol	2.73	61.16	58.73	50.10	55.16	53.12	37.93	42.16	14.66
A4	3-methyl-1-butanol	179	1.31	2.36	1.82	1.28	1.27	2.27	2.03	1.39
A5	1-hexanol	5.37	11.67	4.35	6.85	9.08	10.38	8.27	9.75	11.58
AL1	furfural	0.122	58.48	30.78	70.50	73.71	72.58	44.03	53.94	56.02
AL2	phenylacetaldehyde	262	3.28	2.02	7.28	8.84	5.57	9.01	7.25	2.50
AL3	nonanal	122	1.25	1.42	1.17	1.13	1.14	1.75	1.11	2.84
AL4	(E)-2-nonenal	51	1.35	1.06	<1	<1	<1	<1	<1	<1
E1	ethyl acetate	32.6	14.97	39.03	45.29	44.45	51.47	49.55	51.10	45.89
E2	ethyl butyrate	0.0815	3350.59	3941.74	4421.50	4543.07	4762.31	4576.04	4871.55	3835.23
E3	ethyl lactate	128	5.93	7.17	10.10	6.75	8.28	7.42	9.76	5.73
E4	ethyl isovalerate	0.007	475.71	408.57	204.29	215.71	295.71	442.86	267.14	384.29
E5	ethyl pentanoate	0.027	2596.50	1976.98	2683.99	2983.81	3394.03	4665.72	4015.88	6044.19
E7	ethyl isocaproate	389	1.71	<1	<1	<1	1.47	2.62	1.46	2.80
E8	ethyl hexanoate	0.055	45299.64	31626.18	52218.36	39842.36	43102.73	53428.73	49287.27	60285.27
E9	hexyl acetate	5560	<1	<1	<1	<1	<1	<1	<1	1.80
E13	isoamyl butyrate	20	25.69	16.82	8.59	8.13	9.27	15.56	8.53	24.58
E17	propyl hexanoate	12800	<1	<1	<1	<1	<1	<1	<1	1.01
E18	ethyl heptanoate	13.2	5.84	2.84	3.91	3.11	2.82	5.43	2.98	9.50
E28	butyl hexanoate	5.25	2.06	<1	1.34	<1	<1	1.24	<1	3.00
E29	ethyl octanoate	0.013	5068.62	3823.64	3592.74	2089.94	1805.01	2280.28	1955.27	2819.95
E33	ethyl phenylacetate	407	1.38	<1	4.56	6.00	5.27	4.67	5.82	1.45
E34	isopentyl hexanoate	1400	4.33	2.41	1.78	<1	<1	1.32	<1	2.50
E40	ethyl 3-phenylpropionate	125	10.69	7.88	4.53	3.94	4.30	4.29	4.89	1.79
E42	hexyl hexanoate	1.89	6.01	2.30	3.26	2.70	1.41	2.01	1.99	6.07
E43	ethyl decanoate	1120	<1	1.36	<1	<1	<1	<1	<1	<1
AC1	acetic acid	160	3.08	3.15	2.87	2.97	3.54	3.44	3.18	3.47
AC2	butyric acid	0.964	102.43	142.71	131.85	127.89	170.61	58.14	187.74	114.51
AC3	valeric acid	0.389	45.92	7.55	126.31	68.73	64.36	37.89	38.73	112.56
AC4	hexanoic acid	2.52	131.37	151.22	131.92	129.68	137.83	148.45	127.59	123.92
AC5	heptanoic acid	13.8	<1	<1	<1	<1	1.51	<1	<1	<1
AC6	octanoic acid	2.7	11.02	8.38	1.62	1.80	5.78	5.13	3.60	3.49

JH1 and JH2 from the Jianghuai region contained more long-chain fatty acid esters and branched-chain fatty acid esters, while baijiu from the Sichuan region had more short- and straight-chain fatty acid esters (such as ethyl caproate, ethyl enanthate, and ethyl valerate). The formation of this regional difference might also be related to the different distillation methods. Specifically, the Sichuan region adopts a relatively high distillation temperature, while the Jianghuai region adopts a stepwise temperature-controlled distillation method, which results in distinct ester profiles in baijiu.

Additionally, among the volatile acids, acetic acid, butanoic acid, and hexanoic acid are relatively abundant, contributing to the flavor complexity of the baijiu. 1-Propanol, 2-methyl-1-propanol, 1-butanol, and 3-methyl-1-butanol were the major alcohols in strong-aroma baijiu, which were consistent with other reports that the concentrations of 1-propanol, ethyl acetate, ethyl butyrate, ethyl lactate, ethyl hexanoate, acetic acid and hexanoic acid in the strong-aroma baijiu were more than 100 mg/L, suggesting that these compounds can be preliminarily considered as the volatile framework components of the strong-aroma baijiu.[24] These volatile organic compounds played a significant role in defining the characteristic aroma profile of the strong-aroma baijiu, contributing to its unique sensory attributes.

The PCA was performed to determine whether samples from different regions could be effectively distinguished (Figure 2a). The first principal component (PC1) and the second principal component (PC2) explained 32.4 and 24.2% of the variance, respectively, accounting for 56.6% of the total variance. The baijiu samples from Sichuan were distinctly clustered together although the SC6 sample exhibited some differences from the other SC samples. In contrast, the JH2 sample from Bozhou, Anhui, and the JH1 sample from Suqian, Jiangsu, were clearly separated from the other samples. The PCA results suggested that volatile compounds could effectively differentiate strong-aroma baijiu from different geographical origins, indicating that the terroir has an important impact on the flavor of baijiu.

3.3. OAVs of Volatile Compounds. To explore how different volatile compounds contribute to the aroma of baijiu, we calculated the OAVs for each compound. A higher OAV indicates a greater contribution of that compound to the overall aroma of the baijiu.[25] As indicated in Table 2, a total of 33 aroma compounds were identified as having an odor activity value (OAV) of 1 or higher across all eight samples. These

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Table 3. Concentrations of Nonvolatile Compounds in Different Strong-Aroma Baijiu[a,b]

ID		concentration (μg/L)							
		JH1	JH2	SC1	SC2	SC3	SC4	SC5	SC6
	acids								
NAC1[c]	lactic acid	1040.9b	1359.4ab	632.4c	1063.4b	1508.5a	1407.6ab	1131.4b	504.3c
NAC2	glycolic acid	1435.3a	800.3c	211.4d	259.5d	316.4d	1158.9b	364.1d	420.9d
NAC3	2-hydroxybutyric acid	321.4b	409.2a	72.6f	88.35ef	151.6d	206.2c	117.3de	121.7de
NAC4	levulinic acid	498.6a	213.4bc	1430.cd	30.0e	200.4bcd	255.1b	130.5d	193.0bcd
NAC5	2-furoic acid	1170.2a	520.8c	164.6f	165.4f	337.1d	527.0c	260.3e	875.4b
NAC6	hydracrylic acid	669.0a	439.1bc	342.2c	418.4bc	564.4ab	693.8a	441.7bc	367.6c
NAC7	3-hydroxybutyric acid	425.1b	218.4 cd	245.6 cd	234.4 cd	289.4c	530.0a	231.6 cd	168.5d
NAC8	2-hydroxy-2-methylbutyric acid	548.5a	149.5c	75.6d	77.6d	148.1c	136.8c	115.8c	215.8b
NAC9	3-hydroxyisovaleric acid	323.0a	231.0b	36.9 g	62.7f	106.0e	193.5c	76.2f	131.7d
NAC10	2-hydroxyisocaproic acid	4350.9b	9769.1a	1713.9c	2599.3bc	3419.2bc	4190.7b	2841.3bc	1338.4c
NAC11	benzoic acid	685.5bcd	352.9d	530.7d	612.0 cd	1415.0a	914.9bc	990.5b	493.6d
NAC12	benzeneacetic acid	1166.3b	576.0c	612.7c	650.1c	1118.1b	1519.1a	837.7c	855.5c
NAC13	maleic acid	351.9a	292.2b	44.6de	33.0e	91.3d	54.8de	50.2de	238.0c
NAC14	succinic acid	536.4b	1273.7a	63.9d	68.5d	134.8d	233.4c	83.1d	148.0d
NAC15	methylsuccinic acid	191.7a	139.4b	39.4e	26.2e	81.0d	106.2c	41.9e	6.2f
NAC16	glyceric acid	181.4a	114.7b	20.6e	33.8de	44.2 cd	122.0b	40.5 cd	57.0c
NAC17	fumaric acid	41.5a	20.7c	12.2d	5.7e	14.9d	21.2c	7.7e	27.4b
NAC18	benzenepropanoic acid	1321.06b	1641.0a	268.8e	217.2e	451.2d	737.7c	352.1de	314.0de
NAC19	mandelic acid	25.5a	18.8b	24.1a	25.5a	25.5a	26.0a	27.5a	23.4a
NAC20	malic acid	1274.5a	582.1b	130.9e	97.9e	190.5e	476.9c	126.0e	329.5d
NAC21	hexanedioic acid	0.25b	27.1b	50.7a	3.6b	27.7b	12.3b	13.2b	13.7b
NAC22	3-phenyllactic acid	334.1b	627.4a	32.3c	31.7c	68.4c	72.5c	49.6c	48.4c
NAC23	2-phenyllactic acid	316.1b	599.7a	23.4c	23.0c	58.2c	62.3c	40.4c	38.9c
NAC24	4-hydroxybenzoic acid	7.5b	7.2b	6.0bc	5.0bc	5.8bc	10.7a	6.5bc	4.3c
NAC25	dodecanoic acid	534.0b	1068.7a	172.8e	166.8e	212.2de	413.1c	254.2d	352.5c
NAC26	tartaric acid	678.7a	621.6a	140.1c	543.0ab	601.1a	535.9ab	420.5b	526.3ab
NAC27	suberic acid	20.4de	59.8c	258.3a	23.4de	76.5b	30.3d	14.4e	19.9de
NAC28	vanillic acid	14.5b	11.3c	8.6de	8.1e	10.2 cd	15.9a	8.65de	9.65de
NAC29	azelaic acid	107.9b	115.2b	574.1a	63.2b	120.8b	100.75 ± b	58.9b	50.6b
NAC30	myristic acid	1027.9c	2104.1a	413.8d	409.4d	574.3d	1424.7b	562.5d	873.7c
NAC31	sebacic acid	19.2bc	20.0bc	128.1a	16.4c	25.0b	19.3bc	15.0c	14.4c
NAC32	pentadecanoic acid	231.5b	391.6a	136.2d	130.9d	179.3c	260.7b	176.4c	233.5b
NAC33	undecanedioic acid	ND	ND	33.9	ND	ND	ND	ND	ND
NAC34	palmitoleic acid	780.8bc	996.1a	597.4 cd	455.3d	817.9b	1163.6a	732.9bc	1032.8a
NAC35[c]	palmitic acid	8.3c	14.5a	3.3e	2.5e	7.7 cd	14.3a	6.3d	11.9b
NAC36	heptadecanoic acid	256.1abc	293.0a	182.2d	170.8d	237.2bc	280.5ab	225.3c	298.1a
NAC37	9,12-octadecadienoic acid (Z,Z)	5268.5d	14140.0a	587.7f	382.3f	7245.8b	6271.0c	772.6f	3848.3e
NAC38[c]	oleic acid	4.5b	2.8c	1.6d	1.2d	3.2c	7.3a	3.1c	6.6a
NAC39	stearic acid	1999.1b	2263.7b	966.3c	721.3c	1874.0b	2985.9a	1780.1b	3071.3a
NAC40	arachidic acid	976.0bc	1105.9ab	619.9de	434.8e	800.5 cd	1168.6ab	821.3 cd	1293.1a
NAC41	behenic acid	985.6ab	1112.8a	831.9bc	631.0c	904.6ab	1015.1ab	819.5bc	1149.5a
NAC42	lignoceric acid	7.5ab	10.3a	5.8bc	2.3c	5.8bc	7.7ab	5.2bc	10.6a
	alcohols								
NA1[c]	2,3-butanediol	13.7bc	10.0c	14.6bc	11.9c	19.2a	12.1c	17.2ab	11.1c
NA2[c]	phenylethyl alcohol	12.9	11.1	12.0	11.4	12.6	12.0	12.4	11.6
NA3	glycerol	656.0b	674.9b	2933.1a	772.2b	725.0b	573.9b	712.2b	592.8b
NA4	2-phenyl-1,2-propanediol	334.3b	537.4a	88.9de	88.3de	147.6c	81.5e	121.8d	87.7de
NA5	myo-inositol	9.7a	5.9b	1.9c	4.5b	6.0b	4.7b	5.1b	0.2d
	sugar alcohols								
SA1	erythritol	44.1a	8.7d	9.3d	15.6 cd	17.2c	15.3 cd	13.1 cd	34.4b
SA2	xylitol	20.0a	12.6c	10.5c	20.1a	21.0a	15.7b	19.6a	11.0c
SA3	arabitol	26.0a	20.0b	18.6b	25.4a	26.0a	25.6a	27.0a	17.1b
SA4	ribitol	23.4b	27.3a	15.6d	21.5b	21.7b	20.0bc	17.8 cd	8.4e
SA5	D-mannitol	32.2a	17.6de	16.5e	21.0bc	19.2 cd	22.4b	20.0c	17.3de
SA6	D-sorbitol	24.5a	21.2b	19.8d	21.2b	21.4b	20.8bc	20.4c	19.8d
	carbohydrates								
C1	α-D-xylopyranose	195.0a	190.48b	187.3c	189.6bc	189.5bc	189.7bc	190.7b	191.5b
C2	D-fructose	560.9a	24.0c	14.0c	13.4c	15.0c	51.2b	28.8c	9.0c

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Table 3. continued

ID		concentration (μg/L)							
		JH1	JH2	SC1	SC2	SC3	SC4	SC5	SC6
C3	D-glucose	1120.1a	109.7c	91.2c	75.9c	115.2c	159.4b	170.7b	72.6c
C4	D-trehalose	37.5b	43.5a	22.3d	22.2d	31.3c	28.4c	29.1c	18.9d
C5	sucrose	278.9	ND	ND	ND	ND	ND	ND	ND
	amino acids								
AA1	alanine	ND	ND	ND	ND	ND	ND	173.8	ND
AA2	L-5-oxoproline	70.7b	55.9a	36.9 cd	32.3cde	41.1c	27.2de	25.5de	22.1e
	esters								
MS	monoethyl succinate	704.5b	1580.4a	139.8d	131.7d	240.3 cd	334.9c	159.1d	173.7d

[a]Compounds marked with different letters have significantly different contents (Tukey, $p < 0.05$). [b]The concentrations of compounds are presented as the mean ± SD ($n = 3$). [c]The concentrations of compounds are expressed in units of mg/L.

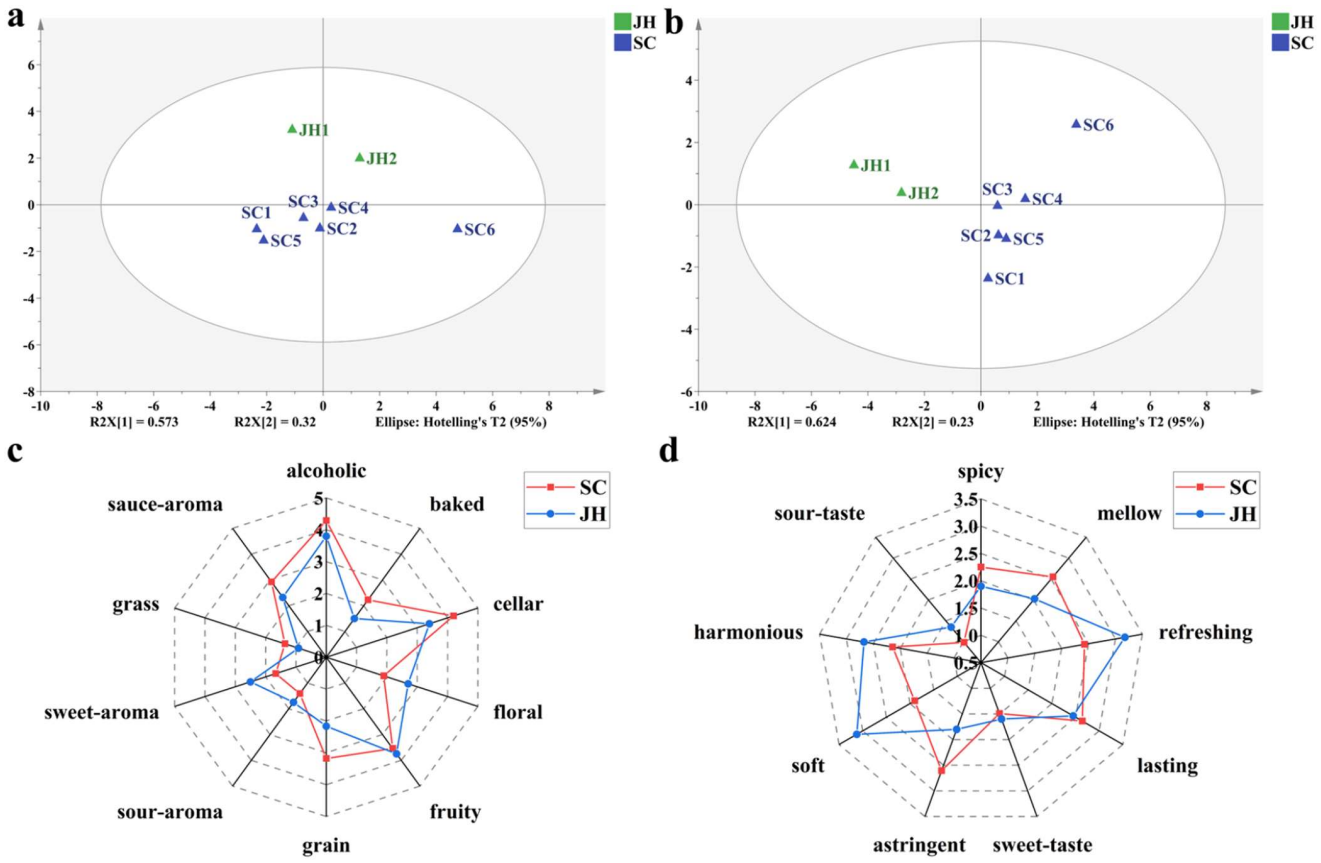


Figure 3. Sensory characters of JH and SC samples by QDA. (a) PCA score shows the aroma differences between JH and SC samples; (b) PCA score shows the taste differences between JH and SC samples; (c) spider chart of aroma descriptors; (d) spider chart of taste descriptors.

include 5 alcohols, 4 aldehydes, 6 acids, and 18 esters. It could be inferred that esters were the most significant aroma compounds in the strong-aroma baijiu, particularly due to their high OAVs, which include ethyl butyrate, ethyl pentanoate, ethyl hexanoate, and ethyl octanoate. Ethyl hexanoate has the highest OAV, particularly in sample SC6, where its OAV exceeded 60000.

Six volatile acids had an OAV of over 1 in at least one sample. Among them, butyric and hexanoic acids had extremely low odor thresholds, resulting in their OAVs exceeding 100 in seven baijiu samples (except SC4). Acids constitute one of the most crucial classes of aromatic components in baijiu, characterized by high content and low perceptual thresholds. For the volatile alcohols in strong-aroma baijiu, the OAV of 1-butanol was the highest, and the OAVs of 1-propanol, 2-methyl-1-propanol, isoamyl alcohol, and 1-hexanol were all greater than 1. The results aligned with previous research that compounds with OAV exceeding 1 in strong-aroma baijiu were esters, acids, and alcohols.[12] The OAV values of the same volatile compounds varied significantly across different strong-aroma baijiu samples, which might be associated with the flavor differences between the samples. Specifically, for the strong-aroma baijiu from the Sichuan region, the contents of compounds such as β-cyclocitral, 2-undecanone, ethyl phenylacetate, phenylacetaldehyde, ethyl butyrate, ethyl acetate, furfural, valeric acid, acetic acid, and ethyl lactate are higher than those of the strong-aroma baijiu from the Jianghuai region. In contrast, for the samples from the Jianghuai region, the contents of compounds such as caprylic acid, ethyl isovalerate, ethyl caprylate, trans-2-

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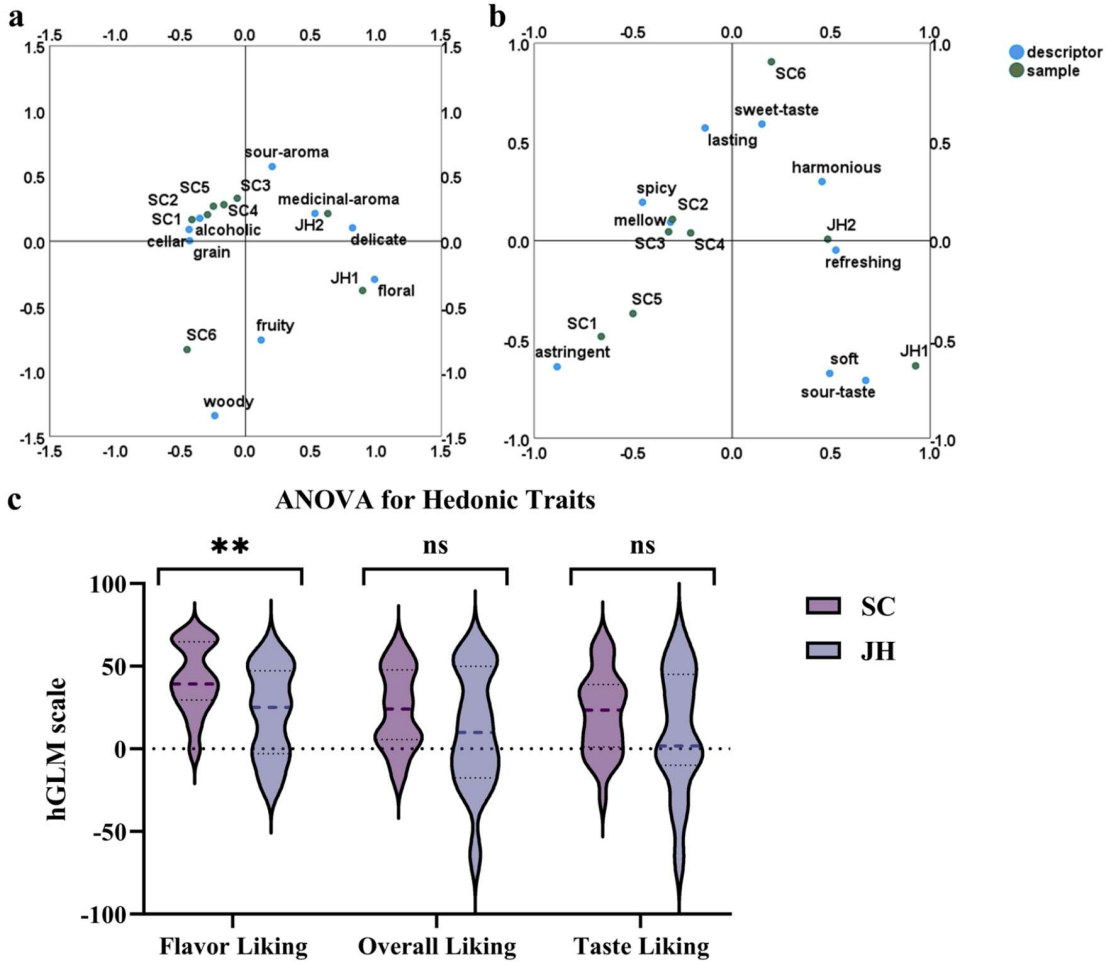


Figure 4. (a) Correspondence analysis of the aroma characteristics of CATA; (b) correspondence analysis of the taste characteristics of CATA; (c) analysis of variance (ANOVA) comparing consumer preference scores between the SC and JH baijiu. **p-value <0.01.

nonenal, ethyl 3-phenylpropionate, isoamyl hexanoate, and isobutanol are significantly higher than those of the Sichuan samples.

3.4. Differences in the Quantity of Nonvolatile Compounds among Various Strong-Aroma Baijiu. A total of 60 nonvolatile compounds were quantified in strong-aroma baijiu (Table 3). Additionally, acids accounted for the highest proportion of nonvolatile compounds in strong-aroma baijiu, followed by alcohols (Figure 1d). The content of nonvolatile acids in Sichuan samples SC1 and SC6 was markedly lower than in the other samples. For the nonvolatile compounds, lactic acid reached a content as high as 1508.54 ± 188.03 mg/L (SC3). Based on the specific quantitative data, the content of nonvolatile organic acids in strong-aroma baijiu exhibits obvious characteristics of regional distribution. The content of trace organic acids (excluding lactic acid) in baijiu from the Sichuan region was less abundant than that from the Jianghuai region. It is noteworthy that baijiu from the Jianghuai region had a significantly higher level of long-chain fatty acids (more than 12 carbon atoms), which should also be related to the different distillation processes. Furthermore, the concentrations of carbohydrates and sugar alcohols were significantly low in strong-aroma baijiu, which could be attributed to the separation in the distillation process. The level of carbohydrates in baijiu from the Jianghuai region was significantly higher than that from the Sichuan region.

Using the acquired quantitative data of nonvolatile compounds, PCA analysis was performed. The first principal component (PC1) and the second principal component (PC2) explained 70.2% of the total variance. As shown in the score plot of Figure 2b, the baijiu samples from the Jianghuai region were primarily located on the negative side of PC2. The PCA showed that the samples could be clearly separated based on their nonvolatile profile. OPLS-DA analysis was performed to identify key nonvolatile compounds that differentiate the Sichuan and Jianghuai samples. The OPLS-DA score plot is shown in Figure 2c. Permutation test (n = 1000) showed that the model was not overfitted (Figure S1, Q^2 = 0.976, R^2Y = 0.986). Twenty-three nonvolatile compounds were identified as potential markers to distinguish between the SC and JH groups (with VIP > 1), including 17 organic acids, 2 alcohols, 1 ester, 2 carbohydrates, and 1 amino acid. Most of the nonvolatile biomarkers were more abundant in JH samples than in the SC group, except for benzoic acid and hexanedioic acid (Figure 2d). Wang et al.[15] reported that the nonvolatile organic acids could be used as characteristic markers to effectively distinguish baijiu from different fermentation processes and different aroma types. These results demonstrated that nonvolatile compounds could serve as discriminative markers for differentiating both the production techniques and regional characteristics of baijiu.

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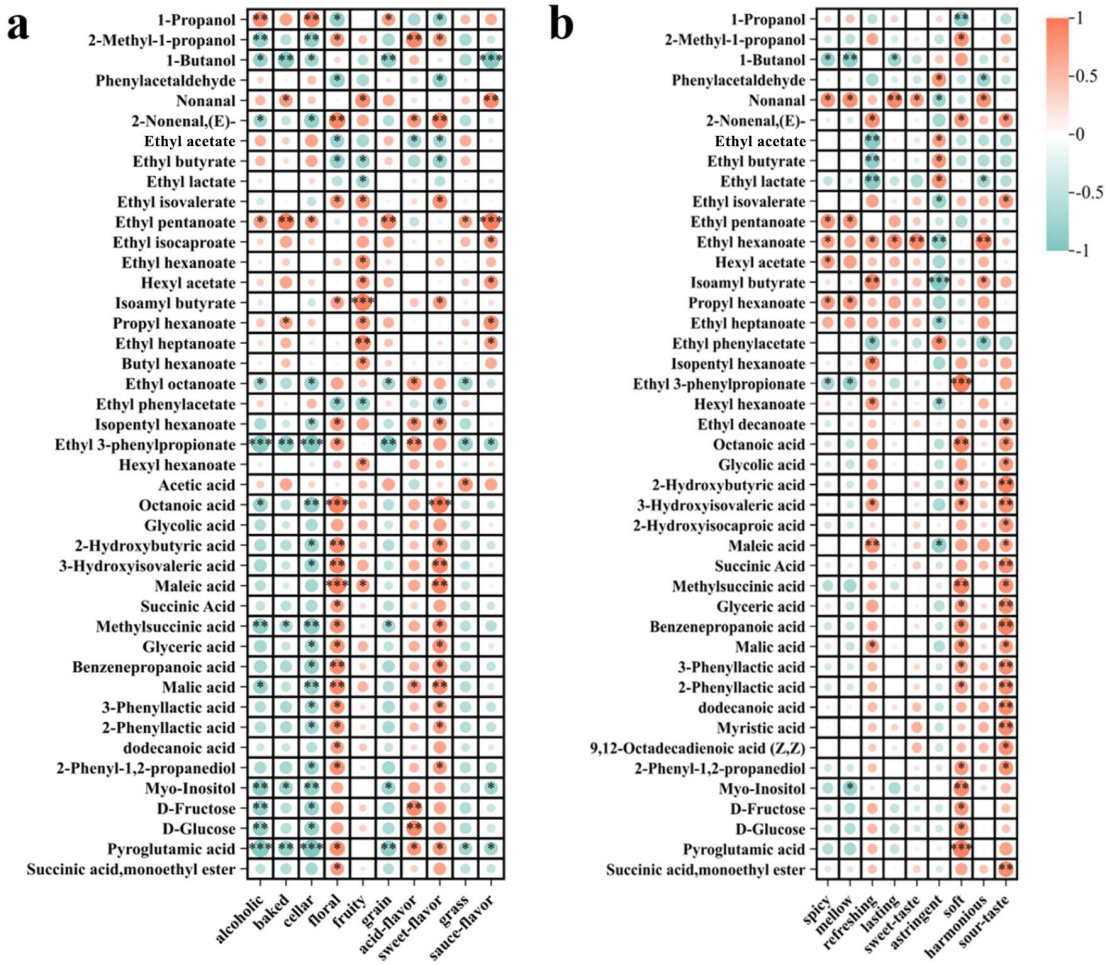


Figure 5. Pearson correlation coefficients between flavor compounds and flavor characteristics based on sensory evaluation. (a) Correlation coefficients between all compounds and aroma descriptors; (b) correlation coefficients between all compounds and taste descriptors. *** p-value <0.001,** p-value <0.01, * p-value <0.05.

3.5. QDA and CATA Analysis. In this study, QDA and CATA were both used to characterize the sensory profile of the baijiu samples to bridge the gap between the expert panel and the consumer panel. First, the QDA results were analyzed by PCA for aroma (orthonasal) and taste/mouthfeel, respectively. The results indicated that samples from different regions could be effectively separated on PC1 using the aroma descriptors (Figure 3a). For taste/mouthfeel descriptors (Figure 3b), samples from different regions could be clearly distinguished on PC2. The QDA indicated that the SC6 sample showed distinct differences in aroma and taste compared with other SC samples, which was consistent with the analytical data. The radar chart (Figure 3c,d) highlights the distinct sensory profiles of the strong-aroma baijiu from different regions. Baijiu from the Jianghuai region displayed significantly higher intensities in aroma characteristics, such as sweetness, acidity, fruitiness, and floral notes. Its taste/mouthfeel was characterized by harmonious, soft, sweet, acidic, and refreshing qualities, particularly when compared with baijiu from the Sichuan region. In contrast, the strong-aroma baijiu from Sichuan was more recognized for its alcoholic, baked, cellar, grainy, and grassy notes. Its taste/mouthfeel characteristics included spiciness, mellowness, flavor lasting, and astringency. The results were consistent with He et al.[10] that the baijiu from the Jianghuai region had a positive correlation with floral, sweet, and fruity aromas, while the baijiu from the Sichuan region had a positive correlation with the aromas of grain, cellar, grass, sour, and baked aroma. However, there were few sensory data on the taste/mouthfeel characteristics of strong-aroma baijiu.

To enhance the understanding of the sensory attributes from consumer perspectives, CATA analysis was further utilized. PCA effectively distinguished the different samples based on the varying frequencies of selected sensory descriptors (Figure 4a). SC6 stood out distinctly from the other baijiu samples, characterized by an intense fruity and woody aroma. The remaining baijiu samples from Sichuan were described by the consumer panel as having alcoholic, cellar, and grain aromas. JH2 was noted for its prominent sour, medicinal, and delicate aromas, while JH1 was characterized by a floral aroma. Additionally, the distances between JH1 and JH2 were relatively close. The differences in taste among the baijiu samples were more pronounced (Figure 4b). SC1 and SC5 were noted for their strong astringency, while SC2, SC3, and SC4 were characterized by spiciness and mellowness. SC6 was recognized for its unique, long-lasting sweetness. JH1 was described as having distinctive sourness and softness, whereas JH2 stood out for its refreshing and balanced flavor.

Although generated by different panels, the results from QDA and CATA were consistent regarding the aroma of baijiu. This indicates that the strong aroma of baijiu from different

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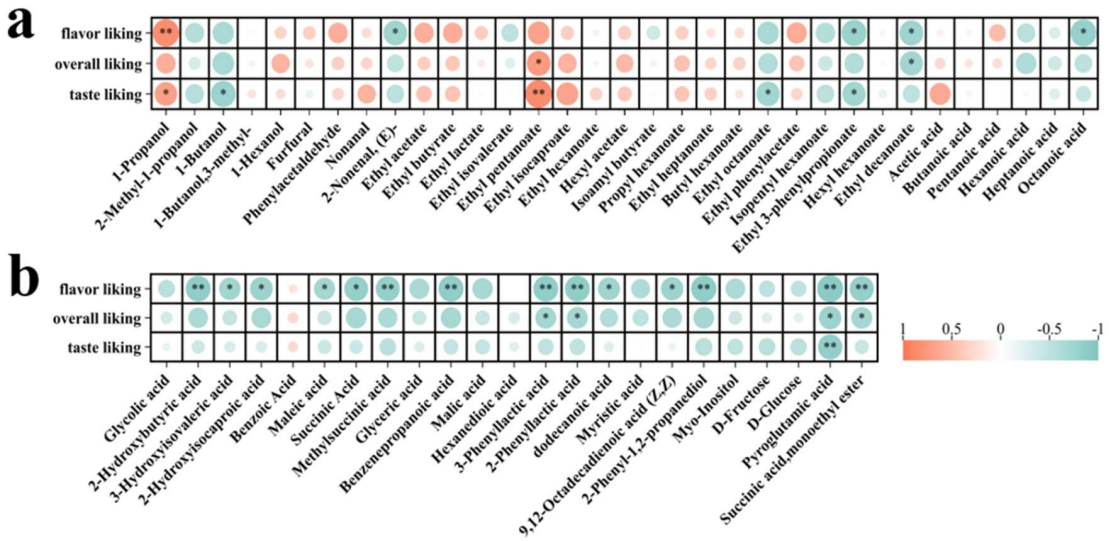


Figure 6. Correlation analysis between the hedonic traits and flavor compounds in strong-aroma baijiu. (a) Pearson correlation between the hedonic traits and volatile compounds; (b) Pearson correlation coefficients between the hedonic traits and nonvolatile compounds. **p-value <0.01, *p-value <0.05.

regions has distinct characteristics that can be easily recognized by consumers. However, when it comes to taste, the CATA could not effectively differentiate the strong-aroma baijiu from various origins. This limitation may arise from the fact that consumers, lacking professional training, often find it hard to discern subtle taste differences between samples.

3.6. Relationships between the Consumer Preference and Baijiu from Different Geographical Origins. Consumer panel provided preference scores for eight strong-aroma baijiu samples from different regions (including flavor, taste, and overall). Figure 4c shows that consumers preferred the aroma of the baijiu from the SC group over that from the JH group. This difference was found to be statistically significant ($p < 0.01$). There was no significant difference in taste/mouthfeel and overall preference between the 2 groups, possibly because the consumer panel could not effectively distinguish the taste/mouthfeel differences between the samples. It was also observed that the score of aroma liking was obviously higher than the score of taste liking, but the latter one was more similar to the score of overall liking. The result indicated that aroma creates a powerful initial impression for strong-aroma baijiu, but the overall liking of the baijiu was more influenced by taste.

Strong-aroma baijiu was described as having an intense aroma, a rich palate, well-coordinated flavors, and a long aftertaste.[1] During CATA testing, the consumer panel demonstrated lexical conservatism in characterizing baijiu samples, predominantly employing heritage sensory descriptors. Strong-aroma baijiu from Sichuan was characterized by a rich aged flavor and a mild soy sauce aroma, while strong-aroma baijiu from Jianghuai is dominated by "cellar" aroma, "qu" (fermentation starter) aroma, and cereal-like aroma, which are less intense than those of Sichuan baijiu. Nevertheless, the oil aging (a tofu-like flavor) was more obvious.[26] In terms of taste, strong-aroma baijiu from Sichuan was characterized by a richer and more harmonious taste, while those from the Jianghuai region were known for their soft and gentle sweetness. Therefore, combining the aforementioned sensory results, it could be inferred that the consumer panel had a preference for the baijiu from the Sichuan region, possibly associated with the stronger aroma intensity, whereas the preferences for the taste/mouthfeel of baijiu varied (Figure 4).

3.7. Relationships between the Sensory Attributes, Flavor Preference, and Trace Compounds. Pearson correlation coefficients were calculated between 56 trace compounds (33 volatile and 23 nonvolatile compounds) and sensory attributes obtained by QDA in order to establish the relationship between trace compounds and sensory properties. Generally, the fruity notes were mainly correlated with the esters, especially isoamyl butyrate and ethyl heptanoate. Alcoholic aroma was positively correlated with 1-propanol. Negative correlations were observed between nonvolatile organic acids and the aroma characteristics of alcoholic and cellar notes in baijiu (Figure 5a). Nonvolatile acids also had certain expression effects on the grain, grass, and sauce flavor. Our results indicated that although nonvolatile organic acids cannot directly affect the aroma of baijiu, they can indirectly influence the volatility of volatile compounds, thus affecting the aroma of baijiu. Furthermore, not all volatile compounds within the same category consistently enhanced a specific aromatic characteristic. This inconsistency might arise from the interactions among different volatile compounds in the complex aroma matrix of baijiu, which could affect their contributions to the overall aromatic profile.

The correlations between trace compounds in baijiu and its taste/mouthfeel characteristics (Figure 5b) indicated that sweetness, which is a key sensory attribute of baijiu, was positively correlated with specific esters and organic acids. For example, ethyl hexanoate was positively correlated to the sweet taste of baijiu, which was consistent with previous research.[27,28] In contrast, the contribution of sugars to sweetness was not as significant as that of esters, which might be due to the concentration of sugars in baijiu being much lower than their taste thresholds. Additionally, it could be observed that the majority of nonvolatile organic acids enhanced the perception of sourness in baijiu. Organic acids such as 2-hydroxybutyric acid, 3-hydroxyisovaleric acid, succinic acid, glyceric acid, benzenepropanoic acid, 3-phenyllactic acid, 2-phenyllactic acid, dodecanoic acid, and myristic acid contributed to the

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sourness most significantly. Our results confirmed a recent study result that both volatile and nonvolatile organic acids played a role in the sour taste of baijiu.[29] Low-chain organic acids are the main contributors to the sour taste in baijiu because they are easily ionized and release hydrogen ions. Medium-chain organic acids help to soften the sharpness of these smaller acids and prolong the sour taste, thus enhancing the overall sourness of baijiu.[30]

Spiciness of baijiu is closely linked to its sensory quality, palatability, and consumers' preferences for baijiu.[31,32] Our results showed that the esters, such as ethyl pentanoate, ethyl hexanoate, propyl hexanoate, hexyl acetate, and nonanal, contributed to the spiciness of baijiu. This observation further confirms the important role of esters in enhancing the spiciness of the spirit, aligning with findings from a previous study by He et al.,[33] which highlighted the key role of esters in the spiciness of distilled spirits.

Additionally, the taste characteristic described as "soft" was closely related to the content of nonvolatile organic acids, indicating they help to balance the palate of the baijiu. On the other hand, esters such as ethyl acetate, ethyl butyrate, and ethyl lactate were negatively correlated with the refreshing mouthfeel. In contrast, ethyl hexanoate and isoamyl butyrate contribute to a more refreshing flavor in baijiu. Furthermore, ethyl hexanoate plays a role in maintaining the lasting aroma of baijiu.

Finally, we investigated the potential links between the hedonic data and the content of trace compounds. The Pearson correlation showed that the volatile compounds, especially 1-pentanol and ethyl pentanoate, had a positive impact on consumers' flavor liking, taste liking, and overall liking (Figure 6). However, ethyl octanoate, ethyl decanoate, and ethyl 3-phenylpropionate were negatively correlated with liking. It is hypothesized that consumer preference for different esters may be influenced by their chain length, as long-chain esters at higher concentrations could potentially exhibit waxy or fatty undertones.

Nonvolatile compounds generally showed a negative effect on liking. Nonvolatile acids could affect the taste of baijiu and indirectly influence the volatility of volatile compounds, thereby impacting the flavor perception of the baijiu.[34] Some studies have pointed out that through molecular dynamics simulation calculations, the attractive intermolecular forces related to ethyl acetate/ethyl caproate increased with oleic acid addition, while those related to ethyl butyrate decreased.[35] Meanwhile, Lyu et al. found that nonvolatile acids lower matrix pH, protonating basic volatiles (e.g., pyrazines) to reduce their volatility while enhancing acidic aromas (e.g., short-chain acids).[36] When the levels of organic acids are too high, they can mask the original aroma of baijiu, resulting in a sour and astringent taste. The organic acids in baijiu could also suppress the characteristic alcoholic and sweet aromas but enhance the fruity, acidic, floral, jujube, and grain aromas,[37] which may also be related to the liking. Our results suggested that nonvolatile acid not only affected the flavor of baijiu but was also related to the consumers' liking.

This study employed a combination of multiple techniques to analyze strong-aroma baijiu in order to explore the relationships among flavor characteristics, trace compounds, and consumer preferences. The content of volatile compounds reveals the differences among strong-aroma baijiu from different regions. Specifically, the alcohol content in baijiu from the Jianghuai region was significantly higher than that in baijiu from the Sichuan region, while the stronger flavor characteristics (such as spicy and overall aroma intensity) of baijiu from the Sichuan region were attributed to its high ester content. Meanwhile, nonvolatile organic acids demonstrated a dual regulatory effect on the aroma-taste balance. On the one hand, there was a negative correlation between nonvolatile acids and the characteristics of alcohol and cellar aroma. On the other hand, they worked in concert with volatile acids to enhance the sour aroma and sour taste and collaborated with specific esters to improve the sweetness of baijiu, forming the taste complexity. Due to the relatively low content of nonvolatile organic acids (excluding lactic acid) in Sichuan samples, it was possible that by reducing the inhibitory effect on aroma characteristics and optimizing the sweet-sour balance, they were more in line with the consumers' liking. Future research could focus on the precise regulation of nonvolatile acids to balance aroma release and taste palatability, so as to enhance the flavor quality of strong-aroma baijiu.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jafc.5c05364.

Methods for identifying volatile compounds and the standard curves utilized in this study (Table S1); the methods for identifying nonvolatile compounds along with their standard curves (Table S2); the definitions and references for the sensory attributes used in the Quantitative Descriptive Analysis (QDA) (Table S3); full quantification data for 84 volatile compounds studied (Table S4); the results of the permutation testing for the OPLS-DA analysis of nonvolatile compounds (Figure S1) (PDF)

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Notes

The authors declare no competing financial interest.

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